United States Patent Application Publication	20250268957
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	Park; Bridgette et al.

Ingestible Nutritive Compositions With Prebiotics, Probiotics, and Postbiotics

Abstract

Compositions including prebiotics, probiotics, and postbiotics for oral administration and methods of using such compositions for improving health of subjects are provided.

Inventors: **Park; Bridgette (Granada Hills, CA), Li; Shuyan (Austin, TX)**

Applicant: **First Day Life, Inc.** (Austin, TX)

Family ID: **1000007803383**

Appl. No.: **18/584809**

Filed: **February 22, 2024**

Publication Classification

Int. Cl.: **A61K35/747** (20150101); **A61K9/00** (20060101); **A61K35/742** (20150101); **A61K36/185** (20060101); **A61P1/00** (20060101); **A61P1/02** (20060101); **A61P15/02** (20060101); **A61P17/00** (20060101)

U.S. Cl.:

CPC **A61K35/747** (20130101); **A61K9/0056** (20130101); **A61K35/742** (20130101); **A61K36/185** (20130101); **A61P1/00** (20180101); **A61P1/02** (20180101); **A61P15/02** (20180101); **A61P17/00** (20180101);

Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention generally relates to nutritional compositions comprising prebiotics, probiotics, and postbiotics for oral administration and methods for using the oral compositions in improving health of subjects (e.g., human subjects).

BRIEF SUMMARY OF THE INVENTION

[0002] In an aspect of the invention, an orally ingestible nutritional composition for improving health is provided. The composition comprises an amount of gold kiwifruit powder, an amount of *Bacillus coagulans*, and an amount of heat-treated *Lactobacillus casei*. The gold kiwifruit powder within the composition may include or consist essentially of Livaux®. The *Bacillus coagulans* within the composition may include or consist essentially of SNZ 1969®. The heat-treated *Lactobacillus casei* within the composition may include or consist essentially of *Lactobacillus casei* 327. The *Lactobacillus casei* 327 may include or consist essentially of FloraSMART®.

[0003] In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 10:1 and about 30:1 gold kiwifruit powder: *Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2:1 and about 15:1 gold kiwifruit powder: *Lactobacillus casei.*

[0004] In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 1.5:1 and about 5:1 *Lactobacillus casei: Bacillus coagulans.*

[0005] In some embodiments, the orally ingestible compositions may take the form of gummies, tablets, capsules, liquids, suspensions, chewables, soft gels, sachets, powders, syrups, liquid suspensions, emulsions, or other solutions. In particular embodiments, the orally ingestible composition is in the form of an individual dosage form. The individual dosage form may take the form of a gummy, a tablet, a capsule, a soft gel, a sachet, or a chew. In particular embodiments, the individual dosage form is a gummy. In further embodiments, an individual oral dosage form may comprise between about 75 mg and 700 mg gold kiwifruit powder; between about 5 mg and 80 mg heat-treated *Lactobacillus casei*; and between about 1 mg and 50 mg *Bacillus coagulans*. In particular embodiments, the individual oral dosage form comprises about 125 mg gold kiwifruit powder; about 25 mg heat-treated *Lactobacillus casei*; and about 7.5 mg *Bacillus coagulans*. In particular embodiments, an individual oral dosage form may comprise an amount of gold kiwifruit powder, an amount of heat-treated *Lactobacillus casei*, and between about 100 million CFUs and 5 billion CFUs *Bacillus coagulans*. In particular embodiments, the individual oral dosage form comprises an amount of gold kiwifruit powder, an amount of heat-treated *Lactobacillus casei*, and about 750 million CFUs *Bacillus coagulans.*

[0006] In an aspect, there is a method for administering orally ingestible compositions to a subject in accordance with the disclosure. In some embodiments, the subject is a mammal. In some embodiments, the subject is a human. In some embodiments, the subject is an adult human. In particular embodiments, the subject is an adult human female. In particular embodiments, the subject is an adult human male. In particular embodiments, the subject is an elderly adult human. In still further embodiments, the subject is an elderly adult female. In still further embodiments, the subject is an elderly adult male. In some embodiments, the subject is a human child under the age of 18. In particular embodiments, the subject is a human child between the ages of 0 and 3. In particular embodiments, the subject is a human child between the ages of 0 and 5. In particular embodiments, the subject is a human child between the ages of 1 and 5. In particular embodiments, the subject is a human child between the ages of 6 and 10. In particular embodiments, the subject is a human child between the ages of 11 and 15. In particular embodiments, the subject is a human child between the ages of 15 and 17. In particular embodiments, the subject is a human teenager between the ages of 13 and 19.

[0007] Compositions according to embodiments herein may be administered a single time, daily on a continual basis, or multiple times per week depending on subject needs, subject body size, and on safety. Dosing may occur once, or only a weekly or monthly basis. Dosage level and frequency of administration protocols, compositions, as well as potential use of additional therapeutic or nutritive agents in combination can be adjusted during the course of administration. In an embodiment, compositions in accordance with the present inventions can be administered daily to a subject. In a particular embodiment, compositions in accordance with the present inventions can be administered multiple times per day to a subject.

[0008] In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day.

[0009] In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day.

[0010] In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.

[0011] In an embodiment, a method of increasing the amount of *Faecalibacterium prausnitzii* within a digestive tract of a subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In particular embodiments, the subject is a human. In particular embodiments, the subject is a human adult. In further embodiments, the subject is a human child under the age of 18. In some embodiments, the subject is a human teenager between the ages of 13 and 19. In some embodiments, the method comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, heat-treated *Lactobacillus casei*, and *Bacillus coagulans* to achieve an increase in the amount of *Faecalibacterium prausnitzii* within the digestive tract of the subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.

[0012] In an embodiment, a method of reducing constipation or relieving symptoms of constipation in a subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In particular embodiments, the subject is a human. In particular embodiments, the subject is a human adult. In further embodiments, the subject is an adult human female. In particular embodiments, the subject is a human child under the age of 18. In some embodiments, the subject is a human teenager between the ages of 13 and 19.

[0013] In some embodiments, the method comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, heat-treated *Lactobacillus casei*, and *Bacillus coagulans* to achieve a reduction in constipation or relief of symptoms of constipation in the subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.

[0014] In some embodiments, the subject's frequency of complete spontaneous bowel movement is increased post-administration of the orally ingestible composition.
[0015] In some embodiments, the subject's stools are softened and/or improved in consistency post-administration of the orally ingestible composition as measured by the Bristol stool form scale.
[0016] In some embodiments, the subject's pain during defecation is reduced post-administration of the orally ingestible composition as measured by the constipation scoring system (CSS) scale.
[0017] In some embodiments, the subject's abdominal pain associated with constipation is reduced post-administration of the orally ingestible composition as measured by the constipation scoring system (CSS) scale.
[0018] In an embodiment, a method of reducing bacterial vaginosis symptoms in a human female subject is provided. The method comprises orally administering, to the human female subject, an orally ingestible composition in accordance with the disclosure herein. In particular embodiments, the subject is an adult human female. In particular embodiments, the subject is a female human child under the age of 18. In some embodiments, the subject is a female human teenager between the ages of 13 and 19. In some embodiments, the method comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, heat-treated *Lactobacillus casei*, and *Bacillus coagulans* to reduce bacterial vaginosis symptoms in the subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day. In some embodiments, the human female subject is undergoing antibiotic treatment for bacterial vaginosis.
[0019] In an embodiment, a method of improving the immune response of a human subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In particular embodiments, the subject is a human child under the age of 18. In particular embodiments, the subject is a human adult. In some embodiments, the subject is a human teenager between the ages of 13 and 19. In some embodiments, the method comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, *Lactobacillus casei*, and *Bacillus coagulans* to improve the immune response of the human subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.
[0020] In some embodiments, incidence frequency of gastrointestinal (GI) infections and/or symptoms associated therewith is decreased in the human subject post-administration of the orally ingestible composition.
[0021] In some embodiments, incidence frequency of upper respiratory tract infections is decreased in the human subject post-administration of the orally ingestible composition.
[0022] In some embodiments, the incidence frequency of colds and cold-like illness in the human subject is decreased post-administration of the orally ingestible composition.

[0023] In an embodiment, a method of improving oral health and reducing tooth decay in a human subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In particular embodiments, the subject is child under the age of 18. In particular embodiments, the subject is a human adult. In further embodiments, the subject is an adult human female. In further embodiments, the human subject is an adult human male. In some embodiments, the subject is a human teenager between the ages of 13 and 19. In some embodiments, the method comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, heat-treated *Lactobacillus casei*, and *Bacillus coagulans* to improve the oral health of and reduce tooth decay in the human subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.
[0024] In some embodiments, salivary *Streptococcus mutans* content is decreased in the oral cavity of the human subject post-administration of the orally ingestible composition in accordance with the disclosure herein. In some embodiments, the incidence of tooth cavities within the oral cavity of the human subject is reduced following oral administration of the orally ingestible composition.
[0025] In some embodiments, the amount of plaque within the oral cavity of the human subject is reduced following oral administration of the orally ingestible composition.
[0026] In some embodiments, the gingival score of the oral cavity of the human subject is reduced following oral administration of the orally ingestible composition.
[0027] In an embodiment, a method of reducing gastrointestinal discomfort in a human subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In particular embodiments, the subject is child under the age of 18. In particular embodiments, the subject is a human adult. In further embodiments, the subject is an adult human female. In further embodiments, the human subject is an adult human male. In some embodiments, the subject is a human teenager between the ages of 13 and 19. In some embodiments, the method comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, heat-treated *Lactobacillus casei*, and *Bacillus coagulans* to reduce gastrointestinal discomfort in the human subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.
[0028] In some embodiments, the human subject's total score on the Severity of Dyspepsia Assessment (SODA) scale is reduced following oral administration of the orally ingestible composition.
[0029] In some embodiments, the human subject's pain intensity subscore on the SODA scale is reduced following oral administration of the orally ingestible composition.

[0030] In some embodiments, the human subject's nonpain symptoms subscore on the SODA scale is reduced following oral administration of the orally ingestible composition.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0031] The foregoing summary, as well as the following detailed description of embodiments of the compositions and methods disclosed herein, will be better understood when read in conjunction with the appended drawings of exemplary embodiments. It should be understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown.

[0032] In the drawings:

[0033] FIG. **1** describes a randomized controlled human trial on the effect of kiwifruit capsules consumptions on gut microflora in functionally constipated individuals. FIG. **1**A lists a description of tested compositions and dosage amounts. FIG. **1**B lists partial results of the study, underscoring the significant increase in abundance of *Faecalibacterium prausnitzii* in functionally constipated individuals with gold kiwifruit powder.

[0034] FIG. **2** summarizes a clinical study on the effect of *L. casei* on constipation symptoms and quality of life. FIG. **2**A displays improvements of abdominal, rectal, and stool symptoms of constipation. FIG. **2**B displays improvement of quality of life symptoms like physical or psychosocial discomfort, worries, concerns, and satisfaction.

[0035] FIG. **3** summarizes a clinically study on the safety and efficacy of *B. coagulans* SNZ 1969 supplementation in reducing infections and symptoms of illness in malnourished children.

[0036] FIG. **4** describes a clinical study of *B. coagulans* SNZ-1969 in the treatment of recurrent bacterial vaginosis (BV). Comparative analysis of the effect of treatment on symptomatic relief (FIG. **4**A) and on Amsels criteria changes (FIG. **4**B) are depicted for both groups.

[0037] FIG. **5** summarizes the effect of *B. coagulans* (*L. sporogenes*) on infantile (FIG. **5**A) and neonatal (FIG. **5**B) diarrhea.

[0038] FIG. **6** summarizes the effect of *B. coagulans* on oral health. Comparison of chlorexhidine, tulsi, and probiotic mouthwash on plaque (FIG. **6**A) and gingival (FIG. **6**B) status; FIG. **6**C: Effect of *B. coagulans*, with or without vitamin B supplementation, on recurrent aphthous ulcerations; FIG. **6**D: effect of *B. coagulans* on salivary *mutans* streptococci compared to control group (left panel) and between groups (right panel). FIG. **6**E: Effect of *B. coagulans* on gingival index (left panel) and plaque index (right panel).

[0039] FIG. **7** Illustrates the effect of *B. coagulans* on gastrointestinal discomfort. FIG. **7**A: Change from baseline to 30 days and day 37 in SODA and GSRS symptoms scores; FIG. **7**B: Frequency of participants experiencing bloating symptom (left panel) or burping symptoms (right panel) in SODA scale at day 30; FIG. **7**C: Frequency of participants experiencing heart burn symptom (left panel) and gas (right panel) in SODA scale at day 30.

[0040] FIG. **8** summarizes the effect of *L. casei* on skin condition. FIG. **8**A displays changes in transepidermal water loss (graphs A and B) and corneum hydration (graphs C and D) measured at weeks 4 and 8 of the trial period. FIG. **8**B displays changes in skin condition measured by a skin questionnaire.

[0041] FIG. **9** summarizes the effect of *L. casei* on colonic serotonin synthesis in mice. FIG. **9**A displays increase in serotonin in the colon tissue as concentration of serotonin (5-HT) and number of 5-HT positive cells. FIG. **9**B displays decrease in colonic transit time though bead expulsion test.

[0042] FIG. **10** summarizes the effect of *L. casei* on defecation through changes in defecation frequency, defecation days, and fecal volume.

DETAILED DESCRIPTION OF THE INVENTION

[0043] Oral supplements and foods including probiotics, beneficial live microorganisms that confer health benefits on their "host" organisms (e.g., humans), have gained popularity as nutritional and health aids. However, to confer a substantial and/or lasting health benefit on those using them, the probiotics must be administered in an adequate amount and have an environment in which they can persist within the subject. It is often difficult to provide an adequate "stabilized" microbiome environment within a "host" organism to reap the full benefits of probiotic supplementation or consumption.

[0044] The interactions between probiotics, prebiotics, and postbiotics have recently garnered attention for the potential to address the issues associated with reaping the benefits of probiotics. Prebiotics may be considered to be probiotic "food"; that is, they are substrates selectively used by probiotic microorganisms in their life cycle maintenance, helping the probiotics survive during digestion to be delivered to the appropriate area of the digestive system. Prebiotics may also be used by the probiotic microorganisms to confer health benefits to the "host" organism. Most, if not, all prebiotic supplements utilize prebiotic fibers like fructooligosaccharides (FOS), xylooligosaccharides (XOS), galacto-oligosaccharides (GOS), or inulin due to low dosage requirements. However, these prebiotics are not considered FODMAP-friendly, as they are fermentable oligosaccharides which can cause unwanted GI side effects (e.g., bloating, gas).

[0045] By contrast, postbiotics may comprise inanimate or dead microorganisms, their respective parts, or waste byproducts produced by probiotic microorganisms that create a healthy microenvironment for probiotic or beneficial microorganisms to flourish. Postbiotics in the form of dead microorganisms may also confer similar health benefits to that of probiotics, but are not at risk of biological instability, since they are already non-viable. It is believed that the interplay between prebiotics, probiotics, and postbiotics may synergistically enhance the beneficial function each is known to confer separately, as the addition of pre- and postbiotics to probiotics is believed to stabilize the life cycle processes of the probiotic microorganisms.

[0046] Here, the inventors have combined pre-, pro-, and postbiotics in surprisingly efficacious manners, yielding beneficial nutritive compositions that demonstrate an enhanced, synergistic effect when consumed by a subject, leading to increased health benefits for the subject. The inventors have also developed surprisingly effective methods of using these beneficial compositions for increasing the health of subjects taking these compositions orally. Moreover, the inventors have used FODMAP-friendly prebiotics in compositions according to the disclosure herein to avoid the issues associated with typical prebiotic-containing foods and supplements.

I. COMPOSITIONS

[0047] In an aspect of the invention, an orally ingestible nutritional composition for improving health is provided. The composition comprises an amount of gold kiwifruit powder (e.g., a prebiotic), an amount of *Bacillus coagulans* (e.g., a probiotic), and an amount of heat-treated *Lactobacillus casei* (e.g., a postbiotic). The gold kiwifruit powder within the composition may include or consist essentially of Livaux®. The *Bacillus coagulans* within the composition may include or consist essentially of SNZ 1969®. The heat-treated *Lactobacillus casei* within the composition may include or consist essentially of *Lactobacillus casei* 327. The *Lactobacillus casei* 327 may include or consist essentially of FloraSMART®.

[0048] In further embodiments, an individual oral dosage form may comprise between about 75 mg and 700 mg gold kiwifruit powder; between about 5 mg and 80 mg heat-treated *Lactobacillus casei*; and between about 1 mg and 50 mg *Bacillus coagulans*. In further embodiments, an individual oral dosage form may comprise between about 75 mg and 100 mg gold kiwifruit powder; between about 5 mg and 15 mg heat-treated *Lactobacillus casei*; and between about 1 mg and 7.5 mg *Bacillus coagulans*. In further embodiments, an individual oral dosage form may comprise between about 100 mg and 200 mg gold kiwifruit powder; between about 15 mg and 25 mg heat-treated *Lactobacillus casei*; and between about 7.5 mg and 12.5 mg *Bacillus coagulans*. In further embodiments, an individual oral dosage form may comprise between about 200 mg and 300 mg gold kiwifruit powder; between about 25 mg and 45 mg heat-treated *Lactobacillus casei*; and between about 12.5 mg and 22.5 mg *Bacillus coagulans*. In further embodiments, an individual oral dosage form may comprise between about 300 mg and 500 mg gold kiwifruit powder; between about 45 mg and 60 mg heat-treated *Lactobacillus casei*; and between about 22.5 mg and 30 mg *Bacillus coagulans*. In further embodiments, an individual oral dosage form may comprise between about 500 mg and 700 mg gold kiwifruit powder; between about 60 mg and 80 mg heat-treated *Lactobacillus casei*; and between about 30 mg and 50 mg *Bacillus coagulans.*

[0049] In particular embodiments, the individual oral dosage form comprises about 75 mg gold kiwifruit powder; about 15 mg heat-treated *Lactobacillus casei*; and about 4.5 mg *Bacillus coagulans*. In particular embodiments, the individual oral dosage form comprises about 125 mg gold kiwifruit powder; about 25 mg heat-treated *Lactobacillus casei*; and about 7.5 mg *Bacillus coagulans*. In particular embodiments, the individual oral dosage form comprises about 250 mg gold kiwifruit powder; about 50 mg heat-treated *Lactobacillus casei*; and about 15 mg *Bacillus coagulans*. In particular embodiments, the individual oral dosage form comprises about 500 mg gold kiwifruit powder; about 100 mg heat-treated *Lactobacillus casei*; and about 30 mg *Bacillus coagulans*. In particular embodiments, the individual oral dosage form comprises about 700 mg gold kiwifruit powder; about 140 mg heat-treated *Lactobacillus casei*; and about 42 mg *Bacillus coagulans.*

[0050] According to certain embodiments of the inventive composition, the compositions are formulated such that they cause improvement in health markers within a subject consuming them (e.g., a demonstrable improvement in digestive health). Health improvements may be measured by improvement in symptoms of particular disorders (e.g., reduction in the amount of *Streptococcus mutans* within the oral cavity), or increase in markers of health (e.g., increase in the amount of *Faecalibacterium prausnitzii* within a digestive tract).

[0051] In a further embodiment, the composition includes concentrations or amounts of the pre-, pro-, and postbiotics that are therapeutically effective to cause improvement in health markers within the subject.

[0052] In a preferred embodiment of the composition, the composition includes gold kiwifruit powder as a prebiotic. Gold kiwifruit powder possesses well-demonstrated tolerability in vivo and is FODMAP-friendly. It is also shown to increase the presence of *Faecalibacterium prausnitzii* in the microbiome, a beneficial gut probiotic microorganism. Gold kiwifruit powder has also been shown to improve gut transit, contribute to fecal bulking, and increase mucus production in the gut. Clinically or therapeutically effective amounts, or amounts sufficient to cause improvement in health markers within the subject, of the gold kiwifruit powder in compositions according to the invention herein include from about 75 mg to about 2800 mg of gold kiwifruit powder per daily total dose, from about 75 mg to about 2500 mg per daily total dose; from about 75 mg to about 2250 mg per daily total dose; from about 75 mg to about 2000 mg per daily total dose; from about 75 mg to about 1750 mg per daily total dose; from about 75 mg to about 1500 mg per daily total dose; from about 75 mg to about 1250 mg per daily total dose; from about 75 mg to about 1000 mg per daily total dose; from about 75 mg to about 750 mg per daily total dose; from about 75 mg to about 500 mg per daily dose; or from about 75 mg to about 250 mg per daily total dose. The gold kiwifruit powder may comprise Livaux®.

[0053] In particular embodiments, clinically or therapeutically effective amounts, or amounts sufficient to cause improvement in health markers within the subject, of the gold kiwifruit powder in compositions according to the invention herein include about 2800 mg of gold kiwifruit powder per daily total dose, about 2500 mg per daily total dose; about 2250 mg per daily total dose; about 2000 mg per daily total dose; about 1750 mg per daily total dose; about 1500 mg per daily total dose; about 1250 mg per daily total dose; about 1000 mg per daily total dose; about 750 mg per daily total dose; about 500 mg per daily dose; about 450 mg per daily total dose; about 400 mg per daily total dose; about 350 mg per daily total dose; about 300 mg per daily total dose; about 250 mg per daily total dose; about 225 mg per daily total dose; about 200 mg per daily total dose; about

175 mg per daily total dose; about 150 mg per daily total dose; about 125 mg per daily total dose; about 100 mg per daily total dose; or about 75 mg per daily total dose.

[0054] In a preferred embodiment of the composition, the composition includes *Bacillus coagulans* as a probiotic. In particularly preferred embodiments, the *B. coagulans* comprises SNZ 1969®. SNZ 1969® has been shown to improve gut, oral, and immune health across multiple therapeutic endpoints and in different human populations, including in children and teenagers.

[0055] Clinically or therapeutically effective amounts, or amounts sufficient to cause improvement in health markers within the subject, of the *Bacillus coagulans* in compositions according to the invention herein include from about 1 mg to about 200 mg of *Bacillus coagulans* per daily total dose, from about 1 mg to about 190 mg of *Bacillus coagulans* per daily total dose, from about 1 mg to about 180 mg of *Bacillus coagulans* per daily total dose, from about 1 mg to about 170 mg of *Bacillus coagulans* per daily total dose, from about 1 mg to about 160 mg of *Bacillus coagulans* per daily total dose, from about 1 mg to about 150 mg per daily total dose; from about 1 mg to about 140 mg per daily total dose; from about 1 mg to about 130 mg per daily total dose; from about 1 mg to about 120 mg per daily total dose; from about 1 mg to about 110 mg per daily total dose; from about 1 mg to about 100 mg per daily total dose; from about 1 mg to about 90 mg per daily total dose; from about 1 mg to about 80 mg per daily total dose; from about 1 mg to about 70 mg per daily total dose; from about 1 mg to about 60 mg per daily total dose; from about 1 mg to about 50 mg per daily total dose; from about 1 mg to about 40 mg per daily total dose; from about 1 mg to about 30 mg per daily total dose; or from about 1 mg to about 20 mg per daily total dose. Clinically or therapeutically effective amounts, or amounts sufficient to cause improvement in health markers within the subject, of the *Bacillus coagulans* in compositions according to the invention herein include between about 100 million CFUs to about 20 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 19 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 18 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 17 billion CFUs *Bacillus coagulans* per daily total dose, about 100 million CFUs and 16 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 15 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 14 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 13 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 12 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 11 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 10 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 9 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 8 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 7 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 6 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 5 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 4 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 3 billion CFUs *Bacillus coagulans* per daily total dose, from about 100 million CFUs to about 2 billion CFUs *Bacillus coagulans* per daily total dose, from about 750 million CFUs to about 10 billion CFUs *Bacillus coagulans* per daily total dose, from about 750 million CFUs to about 5 billion CFUs *Bacillus coagulans* per daily total dose, from about 750 million CFUs to about 2 billion CFUs *Bacillus coagulans* per daily total dose, from about 750 million CFUs to about 1.5 billion CFUs *Bacillus coagulans* per daily total dose, from about 1 billion CFUs to about 15 billion CFUs *Bacillus coagulans* per daily total dose, from about 1 billion CFUs to about 10 billion CFUs *Bacillus coagulans* per daily total dose, from about 1 billion CFUs to about 5 billion CFUs *Bacillus coagulans* per daily total dose, from about 1 billion CFUs to about 2 billion CFUs *Bacillus coagulans* per daily total dose, about 100 million CFUs, about 200 million CFUs, about 300 million CFUs, about 400 million CFUs, about 500 million CFUs, about 750 million CFUS, about 1 billion CFUs, about 1.25 billion CFUs, about 1.5 billion CFUS, about 1.75 billion CFUs, about 2 billion CFUs, about 2.25 billion CFUs, about 2.5 billion CFUS, about 2.75 billion CFUs, about 3 billion CFUs, about 3.25 billion CFUs, about 3.5 billion CFUS, about 3.75 billion CFUs, about 4 billion CFUs, about 4.25 billion CFUs, about 4.5 billion CFUS, about 4.75 billion CFUs, about 5 billion CFUs, about 5.25 billion CFUs, about 5.5 billion CFUS, about 5.75 billion CFUs, about 6 billion CFUs, about 7 billion CFUs, about 8 billion CFUs, about 9 billion CFUs, about 10 billion CFUs, about 11 billion CFUs, about 12 billion CFUs, about 13 billion CFUs, about 14 billion CFUs, about 15 billion CFUs, about 16 billion CFUs, about 17 billion CFUs, about 18 billion CFUs, about 19 billion CFUs, or about 20 billion CFUs per daily total dose. The *B. coagulans* may comprise SNZ 1969®.

[0056] In particular embodiments, clinically or therapeutically effective amounts, or amounts sufficient to cause improvement in health markers within the subject, of the *Bacillus coagulans* in compositions according to the invention herein include about 200 mg of *Bacillus coagulans* per daily total dose, about 190 mg per daily total dose, about 180 mg per daily total dose, about 170 mg of per daily total dose, about 160 mg oper daily total dose, about 150 mg per daily total dose; about 140 mg per daily total dose; about 130 mg per daily total dose; about 120 mg per daily total dose; about 110 mg per daily total dose; about 100 mg per daily total dose; about 90 mg per daily total dose; about 80 mg per daily total dose; about 70 mg per daily total dose; about 60 mg per daily total dose; about 50 mg per daily total dose; about 40 mg per daily total dose; about 30 mg per daily total dose; about 20 mg per daily total dose; about 15 mg per daily total dose; about 10 mg per daily total dose; about 5 mg per daily total dose; about 4 mg per daily total dose; about 3 mg per daily total dose; about 2 mg per daily total dose; or about 1 mg per daily total dose. The *B. coagulans* may comprise SNZ 1969®.

[0057] In a preferred embodiment of the composition, the composition includes heat-treated (i.e., killed) *Lactobacillus casei* as a postbiotic. In particularly preferred embodiments, the heat-treated *L. casei* comprises *Lactobacillus casei* 327. In particular embodiments, the *Lactobacillus casei* 327 may include or consist essentially of FloraSMART®. *Lactobacillus casei* 327 has been shown to support immunomodulatory effects across multiple therapeutic endpoints and in different human populations, as well as improvements in skin condition and hydration.

[0058] Clinically or therapeutically effective amounts, or amounts sufficient to cause beneficial immunomodulatory effects within the subject, of the *L. casei* in compositions according to the invention herein include from about 5 mg to about 320 mg of *L. casei* per daily total dose, from about 5 mg to about 300 mg per daily total dose; from about 5 mg to about 280 mg per daily total dose; from about 5 mg to about 260 mg per daily total dose; from about 5 mg to about 240 mg per daily total dose; from about 5 mg to about 220 mg per daily total dose; from about 5 mg to about 200 mg per daily total dose; from about 5 mg to about 180 mg per daily total dose; from about 5 mg to about 160 mg per daily total dose; from about 5 mg to about 140 mg per daily total dose; from about 5 mg to about 120 mg per daily total dose; from about 5 mg to about 100 mg per daily total dose; from about 5 mg to about 80 mg per daily total dose; from about 40 mg to about 60 mg per daily total dose; or from about 45 mg to about 55 mg per daily total dose.

[0059] In particular embodiments, clinically or therapeutically effective amounts, or amounts sufficient to cause beneficial immunomodulatory effects within the subject, of the *L. casei* in compositions according to the invention herein include about 320 mg of *L. casei* per daily total dose, about 300 mg per daily total dose; about 280 mg per daily total dose; about 260 mg per daily total dose; about 240 mg per daily total dose; about 220 mg per daily total dose; about 200 mg per daily total dose; about 180 mg per daily total dose; about 160 mg per daily total dose; about 140 mg per daily total dose; about 120 mg per daily total dose; about 100 mg per daily total dose; about 80 mg per daily total dose; about 60 mg per daily total dose; about 55 mg per daily total dose; about 50 mg per daily total dose; about 45 mg per daily total dose; about 40 mg per daily total dose; about 35 mg per daily total dose; about 30 mg per daily total dose; about 25 mg per daily total dose; about 20 mg per daily total dose; about 15 mg per daily total dose; about 10 mg per daily total dose; or about 5 mg per daily total dose. The *L. casei* may comprise *Lactobacillus casei* 327. The *Lactobacillus casei* 327 may include or consist essentially of FloraSMART®.

[0060] Compositions according to embodiments of the invention may include combinations of components each in clinically or therapeutically effective amounts. For example, in some compositional embodiments, each of the gold kiwifruit powder, the *B. coagulans*, and the *L. casei* may be included in therapeutically effective amounts in composition.

[0061] In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 10:1 and about 30:1 gold kiwifruit powder: *Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 10:1 and about 25:1 gold kiwifruit powder: *Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 10:1 and about 20:1 gold kiwifruit powder: *Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 15:1 and about 20:1 gold kiwifruit powder: *Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 12:1 and about 18:1 gold kiwifruit powder: *Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 15:1 and about 18:1 gold kiwifruit powder: *Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 22.5:116:1 and about 17:1 gold kiwifruit powder: *Bacillus coagulans*.

[0062] In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2:1 and about 15:1 gold kiwifruit powder: *Lactobacillus casei*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2:1 and about 12.5:1 gold kiwifruit powder: *Lactobacillus casei*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2:1 and about 10:1 gold kiwifruit powder: *Lactobacillus casei*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2:1 and about 7.5:1 gold kiwifruit powder: *Lactobacillus casei*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2:1 and about 5:1 gold kiwifruit powder: *Lactobacillus casei*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 3:1 and about 7:1 gold kiwifruit powder: *Lactobacillus casei*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 4:1 and about 6:1 gold kiwifruit powder: *Lactobacillus casei.*

[0063] In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 1.5:1 and about 5:1 *Lactobacillus casei: Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 1.5:1 and about 4.5:1 *Lactobacillus casei: Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 1.5:1 and about 4:1 *Lactobacillus casei: Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 1.5:1 and about 3.5:1 *Lactobacillus casei: Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2.5:1 and about 4:1 *Lactobacillus casei: Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2:1 and about 4:1 *Lactobacillus casei: Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 2.5:1 and about 3.5:1 *Lactobacillus casei: Bacillus coagulans*. In some embodiments, components of the inventive composition may be combined at a w/w ratio of between about 3:1 and about 3.5:1 *Lactobacillus casei: Bacillus coagulans.*

[0064] Compositions according to certain embodiments herein may be formulated as solid or liquid dosage forms. Dosage forms may be formulated for oral delivery. Oral delivery methods for compositions according to certain embodiments of the invention herein may include gummies, tablets, capsules, liquids, suspensions, chewables, soft gels, sachets, powders, syrups, liquid suspensions, emulsions, or other solutions. In particular embodiments, the orally ingestible composition is in the form of an individual dosage form. The individual dosage form may take the form of a gummy, a tablet, a capsule, or a chew. In particular embodiments, the individual dosage form is a gummy.

II. METHODS

[0065] In an embodiment, a method of increasing the amount of *Faecalibacterium prausnitzii* within a digestive tract of a subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In particular embodiments, the subject is a human. In particular embodiments, the subject is a human child under the age of 18. In particular embodiments, the subject is a human child between the ages of 0 and 3. In particular embodiments, the subject is a human child between the ages of 0 and 5. In particular embodiments, the subject is a human child between the ages of 1 and 5. In particular embodiments, the subject is a human child between the ages of 6 and 10. In particular embodiments, the subject is a human child between the ages of 11 and 15. In particular embodiments, the subject is a human child between the ages of 15 and 17. In particular embodiments, the subject is a human adult. In further embodiments, the subject is an adult human female. In particular embodiments, the subject is a human teenager between the ages of 13 and 19. In some embodiments of increasing the amount of *Faecalibacterium prausnitzii* within a digestive tract of a subject, the method comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, *Lactobacillus casei*, and *Bacillus coagulans* to achieve an increase in the amount of *Faecalibacterium prausnitzii* within the digestive tract of the subject. In some embodiments of the method of increasing the amount of *Faecalibacterium prausnitzii* within a digestive tract of a subject, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.

[0066] In an embodiment, a method of reducing constipation or relieving symptoms of constipation in a subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In some embodiments, the subject is a human. In particular embodiments, the subject is a human child under the age of 18. In particular embodiments, the subject is a human child between the ages of 0 and 3. In particular embodiments, the subject is a human child between the ages of 0 and 5. In particular embodiments, the subject is a human child between the ages of 1 and 5. In particular embodiments, the subject is a human child between the ages of 6 and 10. In particular embodiments, the subject is a human child between the ages of 11 and 15. In particular embodiments, the subject is a human child between the ages of 15 and 17. In particular embodiments, the subject is a human adult. In particular embodiments, the subject is a human teenager between the ages of 13 and 19.

[0067] In some embodiments, the method of reducing constipation or relieving symptoms of constipation in a subject comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, *Lactobacillus casei*, and *Bacillus coagulans* to achieve a reduction in constipation or relief of symptoms of constipation in the subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.

[0068] In some embodiments of the method of reducing constipation or relieving symptoms of constipation in a subject, the subject's frequency of defecation is increased post-administration of the orally ingestible composition. In some embodiments, the subject's frequency of defecation is increased post-administration to at least three bowel movements per week.

[0069] In some embodiments of the method of reducing constipation or relieving symptoms of constipation in a subject, the subject's frequency of complete spontaneous bowel movements is increased. In some embodiments, the subject's frequency of complete spontaneous bowel movements is increased post-administration to at least three complete spontaneous bowel movements per week.

[0070] In some embodiments of the method of reducing constipation or relieving symptoms of constipation in a subject, the subject's stools are softened and/or improved in consistency post-administration of the orally ingestible composition as measured by the Bristol stool form scale. In some embodiments, the subject's stool consistency post-administration achieves a 3 or 4 on the Bristol stool form scale.

[0071] In an embodiment, a method of improving immune health in a subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In some embodiments, the subject is a human. In particular embodiments, the subject is a human child under the age of 18. In particular embodiments, the subject is a human child between the ages of 0 and 3. In particular embodiments, the subject is a human child between the ages of 0 and 5. In particular embodiments, the subject is a human child between the ages of 1 and 5. In particular embodiments, the subject is a human child between the ages of 6 and 10. In particular embodiments, the subject is a human child between the ages of 11 and 15. In particular embodiments, the subject is a human child between the ages of 15 and 17. In particular embodiments, the subject is a human adult. In particular embodiments, the subject is a human teenager between the ages of 13 and 19.

[0072] In some embodiments, the method of improving immune health in a subject comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, *Lactobacillus casei*, and *Bacillus coagulans* to achieve an improvement in immune health of a subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.

[0073] In some embodiments of the method of improving immune health in a subject, the incidence frequency of colds and/or cold symptoms in the subject are decreased post-administration of the orally ingestible composition. In some embodiments, the subject's frequency of experiencing a cold and/or exhibiting cold symptoms is reduced by at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

[0074] In some embodiments of the method of improving immune health in a subject, the incidence frequency of upper respiratory infections and/or symptoms thereof in the subject are decreased post-administration of the orally ingestible composition. In some embodiments, the subject's frequency of experiencing upper respiratory infections and/or symptoms thereof is reduced by at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

[0075] In some embodiments of the method of improving immune health in a subject, the incidence frequency of gastrointestinal (GI) infections and/or symptoms thereof in the subject are decreased post-administration of the orally ingestible composition. In some embodiments, the subject's frequency of experiencing GI infections and/or symptoms thereof is reduced by at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

[0076] In an embodiment, a method of improving oral health in a subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In some embodiments, the subject is a human. In particular embodiments, the subject is a human child under the age of 18. In particular embodiments, the subject is a human child between the ages of 0 and 3. In particular embodiments, the subject is a human child between the ages of 0 and 5. In particular embodiments, the subject is a human child between the ages of 1 and 5. In particular embodiments, the subject is a human child between the ages of 6 and 10. In particular embodiments, the subject is a human child between the ages of 11 and 15. In particular embodiments, the subject is a human child between the ages of 15 and 17. In particular embodiments, the subject is a human adult. In further embodiments, the subject is an adult human female.

[0077] In some embodiments, the method of improving oral health in a subject comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, *Lactobacillus casei*, and *Bacillus coagulans* to achieve an improvement in oral health of a subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.

[0078] In some embodiments of the method of improving oral health in a subject, the amount of *Streptococcus mutans* within the oral cavity of the subject is reduced post-administration of the orally ingestible composition. In some embodiments, the amount of *Streptococcus mutans* within the oral cavity of the subject is reduced by at least 1%, at least 3%, at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

[0079] In some embodiments of the method of improving oral health in a subject, the incidence frequency of tooth cavities within the oral cavity of the subject is decreased post-administration of the orally ingestible composition. In some embodiments, the subject's frequency of experiencing tooth cavities is reduced by at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

[0080] In some embodiments of the method of improving oral health in a subject, the amount of plaque within the oral cavity of the subject is reduced post-administration of the orally ingestible composition. In some embodiments, the amount of plaque within the oral cavity of the subject is reduced by at least 1%, at least 3%, at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

[0081] In some embodiments of the method of improving oral health in a subject, the gingival score of the oral cavity of the subject is decreased post-administration of the orally ingestible composition. In some embodiments, the subject's gingival score is reduced by at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

[0082] In an embodiment, a method of reducing bacterial vaginosis symptoms in a human female subject is provided. The method comprises orally administering, to the human female subject, an orally ingestible composition in accordance with the disclosure herein. In particular embodiments, the subject is a human female child under the age of 18. In particular embodiments, the subject is a human female child between the ages of 0 and 3. In particular embodiments, the subject is a human female child between the ages of 0 and 5. In particular embodiments, the subject is a human female child between the ages of 1 and 5. In particular embodiments, the subject is a human female child between the ages of 6 and 10. In particular embodiments, the subject is a human female child between the ages of 11 and 15. In particular embodiments, the subject is a human female child between the ages of 15 and 17. In particular embodiments, the subject is an adult human female. In some embodiments, the subject is a human female teenager between the ages of 13 and 19

[0083] In some embodiments, the method of reducing bacterial vaginosis symptoms comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, *Lactobacillus casei*, and *Bacillus coagulans* to reduce bacterial vaginosis symptoms in the subject. In some embodiments, the method of reducing bacterial vaginosis symptoms in a human female subject comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, *Lactobacillus casei*, and *Bacillus coagulans* to achieve a reduction in the bacterial vaginosis symptoms of the human female subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day. In some embodiments, the human female subject is undergoing antibiotic treatment for bacterial vaginosis.

[0084] In an embodiment, a method of reducing gastrointestinal discomfort in a subject is provided. The method comprises orally administering, to the subject, an orally ingestible composition in accordance with the disclosure herein. In some embodiments, the subject is a human. In particular embodiments, the subject is a human child under the age of 18. In particular embodiments, the subject is a human child between the ages of 0 and 3. In particular embodiments, the subject is a human child between the ages of 0 and 5. In particular embodiments, the subject is a human child between the ages of 1 and 5. In particular embodiments, the subject is a human child between the ages of 6 and 10. In particular embodiments, the subject is a human child between the ages of 11 and 15. In particular embodiments, the subject is a human child between the ages of 15 and 17. In particular embodiments, the subject is a human adult. In some embodiments, the subject is a human teenager between the ages of 13 and 19.

[0085] In some embodiments, the method of reducing gastrointestinal discomfort in a subject comprises orally delivering to the subject an effective amount or concentration of one or more of gold kiwifruit powder, *Lactobacillus casei*, and *Bacillus coagulans* to achieve a reduction in gastrointestinal discomfort of the subject. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 75 mg and 2800 mg of gold kiwifruit powder per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 5 mg and 320 mg of heat-treated *Lactobacillus casei* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 1 mg and 200 mg of *Bacillus coagulans* per day. In some embodiments, the subject receives, via oral administration of compositions in accordance with the disclosure herein, between about 100 million CFUs and 20 billion CFUs *Bacillus coagulans* per day.

[0086] In some embodiments of the method of reducing gastrointestinal discomfort in a subject, the subject's total score on the Severity of Dyspepsia Assessment (SODA) scale is reduced post-administration of the orally ingestible composition. In some embodiments, the subject's total score on the SODA scale is reduced by at least 1%, at least 3%, at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

[0087] In some embodiments of the method of reducing gastrointestinal discomfort in a subject, the subject's nonpain symptoms subscore on the SODA scale is reduced post-administration of the orally ingestible composition. In some embodiments, the subject's nonpain symptoms subscore on the SODA scale is reduced by at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, or at least 50% post-administration of the orally ingestible composition.

III. EXAMPLES

Example 1: Effect of Gold Kiwifruit Capsules Consumptions on Gut Microflora in Functionally Constipated Individuals

[0088] A randomized controlled human trial demonstrated that consumption of gold kiwifruit capsules (600 mg/capsule, up to 2400 mg/day for 14 days vs. isomalt placebo) significantly increased *Faecalibacterium* bacterial group abundance in functionally constipated subjects (see FIG. **1**, from Blatchford et al., (2017) Consumption of kiwifruit capsules increases *Faecalibacterium prausnitzii* abundance in functionally constipated individuals: a randomised controlled human trial, Journal of Nutritional Science (2017), vol. 6, e52, page 1 of 10, herein incorporated by reference in its entirety). The study design was a randomised double-blind placebo-controlled cross-over trial with participants consuming four different interventions for 4 weeks each, with a 2-week washout between each intervention. The interventions were delivered in 4×600 mg capsules supplied by Anagenix Ltd prepared to look identical to preserve intervention blinding.

[0089] Briefly, participants consumed four different intervention combinations: placebo (isomalt coloured green) (2400 mg/d), ACTAZIN™ L (600 mg/d), ACTAZIN™ H (2400 mg/d) and Livaux™ (2400 mg/d) for 28 d each intervention, with a 14-day washout period between each treatment phase. ACTAZIN™ L (low dose, green kiwifruit) and ACTAZIN™ H (high dose, green kiwifruit) were formulated from cold-processed *Actinidia chinensis* var. *deliciosa* ‘Hayward’ green kiwifruit and Livaux™ was formulated from cold-processed *Actinidia chinensis* var. *chinensis* ‘Zesy002’ gold-fleshed kiwifruit. The placebo ingredient was isomalt (1-O-α-Dglucopyranosyl-D-mannitol). Subjects were asked to exclude high-fibre dietary supplements such as Metamucil, Benefibre and Phloe as well as maintaining their habitual dietary intakes and physical activity habits and to refrain from eating fresh kiwifruit for the study period. At the beginning and end of each 4-week intervention period, participants were asked to provide a faecal sample. The washout period of 2 weeks was chosen to allow sufficient time to return bowel habits to baseline for the parameters measured (microbial ecology, microbial metabolites and SmartPill® pH measurements).

[0090] This study investigated the impact of ACTAZIN™ green (2400 and 600 mg) and Livaux™

(2400 mg) gold kiwifruit supplements on faecal microbial composition and metabolites in healthy and functionally constipated (FC) participants. The participants were recruited into the healthy group (n 20; one of whom did not complete the study) and the FC group (n 9), each of whom consumed all the treatments and a placebo (isomalt) for 4 weeks in a randomised cross-over design interspersed with 2-week washout periods. Modification of faecal microbiota composition and metabolism was determined by 16S rRNA gene sequencing and GC, and colonic pH was calculated using SmartPill® wireless motility capsules. A total of thirty-two taxa were measured at greater than 1% abundance in at least one sample, ten of which differed significantly between the baseline healthy and FC groups. Specifically, *Bacteroidales* and *Roseburia* spp. were significantly more abundant (P<0-05) in the healthy group, and taxa including Ruminococcaceae, *Dorea* spp. and *Akkermansia* spp. were significantly more abundant (P<0-05) in the FC group. In the FC group, *Faecalibacterium prausnitzii* abundance significantly increased (P=0.024) from 3.4 to 7.0% following Livaux™ supplementation, with eight of the nine participants showing a net increase. Lower proportions of *F. prausnitzii* are often associated with gastrointestinal disorders. The discovery that Livaux™ supplementation increased *F. prausnitzii* abundance offers a potential strategy for improving gut microbiota composition, as *F. prausnitzii* is a butyrate producer and has also been shown to exert anti-inflammatory effects in many studies.

Example 2: Effect of Gold Kiwifruit on Bowel Movements

[0091] Participants were administered Livaux in a placebo-controlled study. Participants treated with *L. casei* saw a 162% increase in complete bowel movements as compared to a 35% increase seen in the placebo group. The group administered *L. casei* also saw a statistically significant decrease in abdominal symptoms (e.g. discomfort, pain, bloating and cramps), rectal symptoms (e.g. painful movements, rectal burning, rectal bleeding/tearing), and stool symptoms (e.g. incomplete bowel movement, hard/small bowel movements, straining, squeezing) (FIG. **2**A). The *L. casei* treated group saw a reduction in psychosocial discomfort and increase in satisfaction (FIG. **2**B).

Example 3: Effect of *Bacillus coagulans* on Immunity of Moderately Malnourished Children

[0092] A total of 82 malnourished children aged 1-5 years with evidence of moderate malnutrition (having a weight-for-height z-score of −3.0 to less than −2.0), were enrolled in the study, which was randomized into Group A—probiotic *Bacillus coagulans* SNZ 1969 (n=41) and Group B—Placebo (n=41). Malnourished children participated in this study were randomized in a 1:1 ratio to the probiotic arm (*Bacillus coagulans* SNZ 1969) or placebo arm to receive 5 mL syrup containing 300 million CFU—thrice daily for a period of 3 months. Each 5 mL syrup containing 300 million CFU of *Bacillus coagulans* SNZ 1969 was given thrice daily for a period of 3 months. The study consisted of a baseline visit for screening and randomization followed by follow up visits I, II, & III on day 30, day 60, and day 90, respectively. Subject diaries were used to monitor the compliance of study medications.

[0093] The children experiencing vomiting and diarrhea were markedly lower in the probiotic group as compared to placebo, but statistical significance (p<0.05) was achieved only for “Children experiencing Diarrhea”. Other groups, like “Children experiencing AGI” in the 3rd month & “AGI lasting more than 2 days” also demonstrated a positive trend for probiotic as compared to placebo, but without statistical significance. These findings support the beneficial role of *Bacillus coagulans* SNZ 1969 in improvement of gut immunity. However, there is no difference in the children experiencing Respiratory infections in either group, which could be explained by their malnourishment/short treatment duration of 3 months and the long time taken for subject recruitment across 2-3 seasons in consecutive years.

[0094] An unpaired t-test was performed on one study parameter, “Total days of Illness.” A statistically significant reduction was observed in the total number of days of illness in the group that was administered probiotic *Bacillus coagulans* SNZ 1969 when compared with the placebo group (p=0.009923) (FIG. **3**). This reduction in the total days of illness in the probiotic group as compared to placebo may be attributed to the positive immune modulation of *Bacillus coagulans* SNZ 1969 bolstering the ability to resist a particular disease by improving immunity in malnourished children. See A randomized, double blind, placebo-controlled clinical study to evaluate the efficacy & safety of supplementation with *Bacillus coagulans* SNZ 1969 in reducing infections in malnourished children, herein incorporated by reference in its entirety.

Example 4: Effect of *Bacillus coagulans* SNZ-1969 in the Treatment of Recurrent Bacterial Vaginosis

[0095] The present study aimed to evaluate the efficacy of *Bacillus coagulans* (SNZ-1969) against BV and the ability of this bacterial species to prevent BV recurrence in women aged 18-41 years. The study assessed 173 women, and of these women, 120 showed recurrent BV according to Nugent's and Amsel's criteria. These 120 women were randomly assigned to the following study arms: metronidazole arm (400 mg BID×7 days) and metronidazole (400 mg BID×7 days)+*B. coagulans* arm (120 million CFU/tablet×TID×21 days).

[0096] Symptomatic relief and classification according to Amsel's criteria after treatment on the basis of the treatment regimen and characteristics are shown in FIGS. **4**A and B, respectively. The metronidazole+*B. coagulans* arm showed better success (86.6%) in treating and minimizing recurrence of vaginal infections and, consequently, symptoms associated with BV. See Rani et al., *The Efficacy of Probiotic B. Coagulans* (*snz*-1969) *Tablets in the Treatment of Recurrent Bacterial Vaginosis*, International Journal of Probiotics and Prebiotics Vol. 12, No. 4, pp. 175-182, 2017, herein incorporated by reference in its entirety).

Example 5: Effect of *Bacillus coagulans* (*Lactobacillus sporogenes*) in Neonatal and Infantile Diarrhea

[0097] One hundred and twelve newborn infants in rural India were randomized to receive a daily oral dose of 100 million *Lactobacillus sporogenes* (SNZ 1969) or a placebo for one year. Morbidity was monitored each week for 12 months. Ninety-four (84%) experienced diarrhea due to rotavirus infection. The group fed *L. sporogenes* had fewer episodes of diarrhea (3.4±1.0 *L. sporogenes* group vs 8.6±1.7 in the placebo group, $p<0.02$) and less number of days of illness (13±3 days *L. sporogenes* group vs 35±5 days in the placebo group, $p<0.01$). The episodes of diarrhea were also of shorter duration (3.6±1.0 days *L. sporogenes* group vs 6.8±1.1 days in the placebo group, $p<0.05$). The number of infants who presented with mild to moderate dehydration was 11 in the treated group and 15 in the placebo group; the difference was statistically not significant ($p<0.05$). There was a trend for body weight at one year to be higher (10.2±0.2) in the treated group compared with infants who were controls (9.8±0.3) but the difference was statistically not significant ($p<0.05$). These observations suggest that the prophylactic feeding of *Lactobacillus* has a preventive effect on the incidence and duration of acute rotavirus diarrhea (FIG. **5**A). See Chandra, Effect of *Lactobacillus* on the incidence and severity of acute rotavirus diarrhoea in infants. A prospective placebo-controlled double-blind study, Nutrition Research 22 (2002) 65-69, herein incorporated by reference in its entirety.

[0098] Diarrhea in newborn babies is also common. After the start of feeding following birth, the frequency of stool increases. The consistency of stool also changes usually on fifth day—known as ‘transitional stool’. One of the factors for this increase in frequency and change in consistency of stool is that the bacterial flora is not well established.

[0099] *Lactobacillus sporogenes/Bacillus Coagulans* SNZ 1969 (in the form of Sporlac tablets) was evaluated for its efficacy in stopping neonatal diarrhea. Cases were selected at random from the maternity ward of the Sassoon General Hospital, Pune for a total of 66. The majority of cases (60) experienced neonatal diarrhea. Only babies having more than 6 watery stools per day were selected for the trial. These babies were thoroughly examined for cord infection, pyoderma, and lung infection. They were also examined for jaundice and associated complaints. The feeding history was also noted.

[0100] A dose of about 5 million *B. coagulans* spores per kg body weight was explored. As the weight of a newborn baby is around 2 to 3 kg, only a ¼ (quarter) tablet of Sporlac, was given once a day to each newborn (each Sporlac tablet contains approximately 60 million spores). The results (shown in FIG. **5**B) indicate that the *B. coagulans* SNZ 1969 was safe and had a high success rate in treating neonatal diarrhea; the overall success rate was 81.7% experienced at least an improvement in diarrhea symptoms, and all newborns experiencing constipation and of jaundice were treated/cured (100%). The average duration for complete recovery was 1.8 days, and dehydration was significantly prevented. See Dhongade and Anjaneyulu, *Sporlac in Neonatal Diarrhoea*, Maharashtra Medical Journal, Vol. XXIII, No. 11, February 1977, herein incorporated by reference in its entirety.

Example 6: Effect of *Bacillus coagulans* on Oral Health

[0101] Probiotics and tulsi extract mouth rinses are a breakthrough approach to preventing plaque and gingivitis through their different defense mechanisms. Comparison of the effectiveness of chlorhexidine (CHX, 0.2%), tulsi extract (1 g Tulsi extract, 3 g Tween 80, 0.2 g menthol, 3 g sorbitol in 100 mL of water) and probiotic mouth rinses (1 g powder of 150 million spores of *bacillus Coagulans* in 20 mL of water) among 12-15-year-old school children of Mysore, India were studied and compared in a triple-blinded, randomized, and controlled clinical trial-concurrent parallel design. The study was done in a school which was selected by the lottery method. A total of 60 children who fulfilled the eligibility criteria were randomly selected for the study. After baseline examination, subjects were divided and assigned by computer generated table into three groups—Group A: 0.2% CHX mouth rinse, Group B: tulsi extract mouth rinse, and Group C: probiotic (*B. coagulans*) mouth rinse. There was a significant reduction in plaque and gingivitis among all three study products ($P<0.05$). Compared to baseline data, plaque and gingivitis reduction was more pronounced in probiotic mouth rinse compared to chlorhexidine and tulsi extract mouth rinses (FIG. **6**A, **6**B). All the study products, i.e., CHX, tulsi extract, and probiotic mouth rinses showed a significant reduction in plaque and gingivitis compared to baseline scores. See Manjunathappa et al., *Effect of Tulsi Extract and Probiotic Mouthrinse on Plaque and Gingivitis among School Children—A Randomized Controlled Trial*, Journal of Datta Meghe Institute of Medical Sciences University, Vol. 15, Issue 1, January-March (2020), herein incorporated by reference in its entirety.

[0102] Additionally, about one in five suffer from recurrent aphthous ulceration in their life, with little difference in incidence among the two sexes. No causative agent has been identified to date, but some risk factors have been suggested, such as vitamin B-Complex deficiency, emotional and environmental stress, food or drug allergy, endocrine imbalance, and blood dyscrasias. *Lactobacillus sporogenes* (*B. coagulans*) helps the restoration of vitamin B-Complex deficiency, and was selected for study as a potential treatment of aphthous ulcerations. A total of 150 patients suffering from recurrent aphthous ulcerations were included in the trials, irrespective of their age, sex, and race, from the outpatient sections of the Departments of Dental Surgery and Medicine of the Nehru Hospital associated with BRD Medical College Gorakhpur. Patients having underlying etiological factors other than vitamin B deficiency were excluded. Subjects were divided in 4 groups, and given supportive treatment (Diazepam 5 mg/tablet, twice a day for 5 days) if and when thought appropriate (sub group D): [0103] Main group "A": 50 patients treated with Sporlac (60 million spores of *L. sporogenes*/tablet, 2 tablets/day for 5 days)+B-complex vitamin therapy [0104] Main group "B": 50 patients B-complex vitamin therapy only [0105] Main group "C": 50 patients treated with Sporlac only [0106] Sub group A-1: 39 patients derived from failure of Main groups "B" and "C" that were later on treated as in Main group "A" [0107] Sub group "D": 31 patients derived from failure of main group "A" and sub-group of Main group "A-1" that were later on treated with diazepam, B-complex and Sporlac. Maximum cure rate was observed in Group A (86%); then in group C (58%), and the group B (18%).

[0108] Subgroup A-1 showed a cure rate of 54.7%, while subgroup D had a cure rate of 58% (FIG. **6**C). See Sharma et al., Clinical trial of Sporlac in the treatment of recurrent aphthous ulceration, U.P. State Dental Journal Vol. 11, January (1980), 7-12, herein incorporated by reference in its

entirety.
[0109] Another study was carried out to evaluate the effect of *B. coagulans* on salivary *mutans* streptococci counts in children and to explore its anti-caries potential. A placebo-controlled study was undertaken with 3 parallel arms comprising a total of 150 healthy children (7-14 years). The subjects were randomly divided in the groups (each comprising 50 children): group A: placebo powder; group B: a freeze-dried powdered combination of *Lactobacillus rhamnosus* and *Bifidobacterium* species (Darolac (Aristo Pharmaceuticals), 1 g powder of 1.25 billion freeze-dried bacterial combination); group C: a freeze-dried powdered preparation of *Bacillus coagulans* (Sporolac (Uni-Sankyo Ltd) 1 g powder with 150 million spores of *B. coagulans*). Subjects were instructed to mix the preparation in 20 mL of water and to follow a swish and swallow method for 14 days. *Mutans* streptococci colony counts per ml of saliva were performed on *Mitis-Salivarius* Bacitracin agar collected on the first day and 14 days post-intervention. A statistically significant reduction ($p<0.001$) in salivary *mutans* streptococci counts was recorded in both groups B and C after 14 days of probiotic ingestion. In conclusion, a cost-effective probiotic such as *B. coagulans* might be a subject for further research for prevention of caries in children (FIG. **6**D). See Jindal et al., A comparative evaluation of probiotics on salivary *mutans* streptococci counts in Indian children, European Archives of Paediatric Dentistry, 12(4) (2011), herein incorporated by reference in its entirety).
[0110] Another study aimed to evaluate the effectiveness of freeze-dried powdered *B. coagulans* on gingival status and plaque inhibition among 12-15-year-old schoolchildren. This randomized controlled trial was conducted among 12-15-year-old schoolchildren in Jaipur. Commercially available freeze-dried probiotics containing *Lactobacillus acidophilus, Bifidobacterium longum, Bifidobacterium bifidum* and *Bifidobacterium lactis* (Prowel, Alkem Laboratories), lactic acid *bacillus* only (Sporolac, Sangyo), and a placebo powder calcium carbonate 250 g (Calcium Sandoz, Novartis) were assigned to two intervention groups and a placebo group each comprising 11 school children. All subjects were instructed to mix the powder in 30 ml of water and swish once daily for 3 min, for 3 weeks. Periodontal clinical parameters were assessed by examining the subjects for Turesky-Gilmore-Glickman plaque index (PI) (Modification of Quigley-Hein PI) and gingival index at baseline, 7th day, 14th day, and 21st day. For both the probiotic groups, a statistically significant reduction ($P<0.05$) in gingival status and plaque inhibition was recorded up to 2nd week of probiotic ingestion. However, no significant difference was observed in the placebo group. The use of probiotic mouth rinses improves the oral health in children by significantly reducing the plaque and gingival scores (FIG. **6**E). See Yousuf et al., *Effect of Freeze-Dried Powdered Probiotics on Gingival Status and Plaque Inhibition: A Randomized, Double-blind, Parallel Study*, Contemporary Clinical Dentistry Vol. 8(1)(2017), herein incorporated by reference in its entirety.
Example 7: Effect of *Bacillus coagulans* on Gastrointestinal Discomfort
[0111] *Bacillus coagulans*-based probiotics are believed to restore gut microbiota and alleviate symptoms of gastrointestinal (GI) discomfort. This study evaluated the efficacy and safety of SNZ 1969 in individuals with GI discomfort. This was a single-center, randomized, placebo-controlled, parallel-arm, double-blind study. Participants with GI discomfort (n=30 in each arm) without a specific pathology were randomized to receive *B. coagulans*-SNZ 1969, TriBac, or placebo, once daily after a major meal, for 30 days. Symptoms were assessed using the Severity of Dyspepsia Assessment (SODA) scale, Gastrointestinal Symptom Rating Scale (GSRS), and Short Form **36** (SF-36) at baseline, day 15, day 30, and 7 days after the end of treatment. A total of 29 participants from SNZ 1969 and 28 from the placebo group completed the study.
[0112] The reduction in total SODA score from baseline at day 30 was significantly higher for SNZ 1969 compared with placebo (18.34 ± 5.35 vs. 12.60 ± 4.79; $p<0.001$) (FIG. **7**A). The mean baseline and day 30 SODA pain intensity subscores for SNZ 1969 were 29.34 ± 1.57 and 13.93 ± 4.42, respectively. The SODA pain intensity subscore reduction from baseline at day 30 was significantly higher for SNZ 1969 arm than in the placebo arm (15.41 ± 4.98 vs. 10.71 ± 3.68; $p<0.001$). The

SODA nonpain symptom subscore reduction reflected relief from burping/belching, heartburn, bloating, passing of gas, sour taste, nausea, and bad breath. The reduction from baseline was significantly more in participants treated with SNZ 1969 compared with placebo at day 30 (7.28 ± 2.23 vs. 4.89 ± 2.94; $p<0.001$). Additionally, among the nonpain symptoms, a significantly greater reduction in sour taste was seen after treatment with SNZ 1969 than placebo 1.52 ± 0.78 vs. 0.75 ± 0.89; $p=0.001$). The SODA satisfaction subscore with abdominal discomfort significantly improved with SNZ 1969 treatment than placebo (-4.34 ± 1.81 vs. -3.00 ± 1.22; $p=0.002$). Significant improvement in SODA scale scores at day 30 compared with placebo was sustained after 7 days of discontinuing the SNZ 1969 treatment ($p<0.05$). These included total SODA score (15.57 ± 3.76 vs. 12.18 ± 4.61; $p=0.003$), SODA pain intensity subscore (15.86 ± 4.20 vs. 10.43 ± 3.97; $p<0.001$), SODA nonpain subscore (8.76 ± 1.99 vs. 5.00 ± 3.01; $p<0.001$), SODA satisfaction score (-5.31 ± 2.49 vs. -3.25 ± 2.37; $p=0.002$), and SODA sour taste (1.52 ± 0.78 vs. 0.71 ± 0.90; $p<0.001$). SNZ 1969 was found to be safe and effective in reducing GI discomfort, especially dyspepsia (FIG. **7**). See Soman et al., Efficacy and Safety of Probiotic *Bacillus coagulans-SNZ* 1969 *in Gastrointestinal Discomfort: A Randomized, Placebo-Controlled Study*, International Journal of Health Sciences and Research Vol. 12; Issue: 3; March 2022, herein incorporated by reference in its entirety.

Example 8: Effect of *L. casei* on Skin Conditions

[0113] Participants were administered heat-killed *L. casei* in a double-blind, placebo-controlled, parallel-group trial. Compared to pretrial levels, the active group experienced a lower level of transepidermal water loss (TEWL) measured at the arm and reduced skin flakiness when compared to the placebo group.

[0114] Subjects were divided into an active food group (Group A, n=32) which consumed 50 mg *L. casei* (approximately 1×10.sup.11 bacteria) daily for eight weeks and a placebo group (Group P, n=32). Subjects were selected from healthy women aged 20-64 years old who had relatively high rates of TEWL and low stool frequency. Clinical surveys for TEWL measurements, stratum hydration, and skin questionnaire were administered at four and eight weeks. Consumption of *L. casei* resulted in a statistically significant reduction of TEWL at the arm at week 4 ($P=0.021$) but not at the cheek (FIG. **8**A). Intake of heat-killed *L. casei* is effective at improving skin condition (FIG. **8**B). Participants consuming *L. casei* saw a statistically significant reduction of skin flakiness at week 4 ($P<0.01$). See Siato er al., *Effects of intake of Lactobacillus casei subp. casei* 327 *on skin conditions: a randomized, double-blind, placebo-controlled, parallel-group study in women*, Bioscience of Microbiota, Food and Health, Vol. 36; Issue: 3; 2017; pp. 111-120, herein incorporated by reference.

Example 9: Effect of *L. casei* on Colonic Production of Serotonin

[0115] Mice were administered *L. casei* in a controlled study to measure the affect of *L. casei* on colonic serotonin production. The mice administered *L. casei* had significantly higher amounts of serotonin (5-HT) than the control mice. *L. casei* can promote colonic 5-HT production and improve GI motor function.

[0116] Mice were divided into two groups. One group (n=6) were administered *L. casei* in sterile deionized water. The control group (n=6) were administered sterile deionized water alone. The colon tissue of the mice was quantitatively and qualitatively analyzed for 5-HT amount. The mice's peristaltic reflexes in the colon and colonic propulsive motility were measured via a bead expulsion test. The mice administered heat-killed *L. casei* had increased 5-HT production in their colons as evidenced by amount of 5-HT in their tissues ($P<0.001$) and the number of 5-HT positive cells ($P<0.001$) (FIG. **9**A). 5-HT has been found to initiate peristaltic reflexes in the colon. The mice treated with *L. casei* saw faster colonic transit times in the bead expulsion test ($P<0.01$) (FIG. **9**B). See Hara, et al., *Heat-killed Lactobacillus casei subsp. casei* 327 *promotes colonic serotonin synthesis in mice*, Journal of Functional Foods, Vol. 47; 2018; pp. 585-589, herein incorporated by reference.

Example 10: Effect of *L. casei* on Fecal Volume

[0117] Subjects were administered *L. casei* in a randomized, double-blind, placebo-controlled, parallel-group study to test the ability of *L. casei* to improve defecation. The group treated with *L. casei* saw increases in defecation frequency, the number of defecation days, and fecal volume compared to the control group.

[0118] Subjects aged 20-64 years old and with relatively low defecation frequencies were randomly assigned to a group treated with *L. casei* (Group A, n=52) and a placebo group (group P, n=52). Subjects kept a log of their stool status throughout the study period. The group treated with *L. casei* saw statistically significant increases in defecation frequency, number of defecation days, and fecal volume when compared to pretrial amounts (P<0.05) (FIG. **10**). See Saito, et al., *Effects of heat-killed Lactobacillus casei* subsp. *casei* 327 *intake on defecation in healthy volunteers: a randomized, double-blind, placebo-controlled, parallel-group study*, Bioscience of Microbiota, Food and Health, Vol. 37; Issue: 3; 2017; pp. 59-65, herein incorporated by reference.

[0119] It will be appreciated by those skilled in the art that changes could be made to the exemplary embodiments shown and described above without departing from the broad inventive concepts thereof. It is understood, therefore, that this invention is not limited to the exemplary embodiments shown and described, but it is intended to cover modifications within the spirit and scope of the present invention as defined by the claims. For example, specific features of the exemplary embodiments may or may not be part of the claimed invention and various features of the disclosed embodiments may be combined. Unless specifically set forth herein, the terms “a” “an” and “the” are not limited to one element but instead should be read as meaning “at least one”.

[0120] Further, to the extent that the methods of the present invention do not rely on the particular order of steps set forth herein, the particular order of the steps should not be construed as limitation on the claims. Any claims directed to the methods of the present invention should not be limited to the performance of their steps in the order written, and one skilled in the art can readily appreciate that the steps may be varied and still remain within the spirit and scope of the present invention.

Claims

1. An orally ingestible composition comprising: an amount of gold kiwifruit powder, an amount of *Bacillus coagulans*, and an amount of heat-treated *Lactobacillus casei.*

2. The composition according to claim 1, wherein the gold kiwifruit powder comprises Livaux®.

3. The composition according to claim 1, wherein the *Bacillus coagulans* comprises SNZ 1969®.

4. The composition according to claim 1, wherein the *Lactobacillus casei* comprises *Lactobacillus casei* 327.

5. The composition according to claim 1, wherein the *Lactobacillus casei* 327 comprises FloraSMART®.

6. The composition according to claim 1, wherein the ratio of the amount of gold kiwifruit powder:the amount of *Bacillus coagulans* is between 10:1 and 30:1.

7. The composition according to claim 1, wherein the ratio of the amount of gold kiwifruit powder:the amount of *Lactobacillus casei* is between 2:1 and 15:1.

8. The composition according to claim 1, wherein the ratio of the amount of *Lactobacillus casei*:the amount of *Bacillus coagulans* is between 1.5:1 and 5:1.

9. The composition according to claim 1, wherein the orally ingestible composition is in the form of a gummy, a tablet, a capsule, a liquid, a suspension, a chewable, a soft gel, a sachet, a powder, a syrup, a liquid suspension, an emulsion, or other solution.

10. The composition according to claim 1, wherein the orally ingestible composition is in the form of an individual oral dosage form.

11. The composition according to claim 1, wherein the individual oral dosage form is a gummy.

12. The composition according to claim 1, wherein the oral dosage form comprises between 75 mg and 700 mg gold kiwifruit powder; between 5 mg and 80 mg heat-treated *Lactobacillus casei*; and between 1 mg and 50 mg *Bacillus coagulans.*

13. The composition according to claim 1, wherein the individual oral dosage form comprises between about 100 million CFUs and 5 billion CFUs *Bacillus coagulans.*

14. The composition according to claim 1, wherein the oral dosage form comprises about 125 mg gold kiwifruit powder; about 25 mg *Lactobacillus casei*; and about 7.5 mg *Bacillus coagulans.*

15. The composition according to claim 1, wherein the oral dosage form comprises about 750 million CFUs *Bacillus coagulans.*

16. A method of increasing the amount of *Faecalibacterium prausnitzii* within a digestive tract of a human subject, wherein the method comprises orally administering, to the human subject, an orally ingestible composition according to claim 1.

17. A method of reducing constipation, reducing gastrointestinal discomfort, or relieving symptoms of constipation in a human subject, wherein the method comprises orally administering, to the human subject, an orally ingestible composition according to claim 1.

18.-**22**. (canceled)

23. A method of reducing transepidermal water loss (TEWL) or skin flakiness in a human subject, wherein the method comprises orally administering, to the human subject, an orally ingestible composition according to claim 1.

24.-**25**. (canceled)

26. A method of increasing fecal volume or colonic production of serotonin in a human subject, wherein the method comprises orally administering, to the human subject, an orally ingestible composition according to claim 1.

27.-**29**. (canceled)

30. A method of improving oral health or immune health of a human subject, or of reducing bacterial vaginosis symptoms in a human female subject, wherein the method comprises orally administering, to the human subject, an orally ingestible composition according to claim 1.

31.-**43**. (canceled)